Coffee Shops vs. Political Party

Presentation

Presented by **Team B8**

Agenda

Research Question

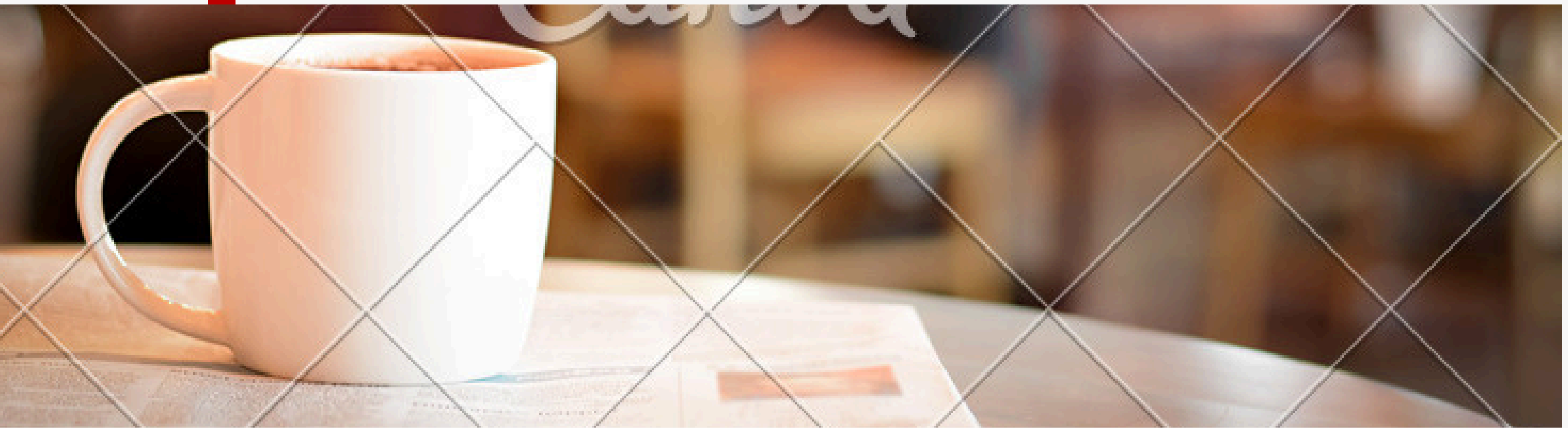
Ideal Experiment

Descriptive Stats
& Visualization

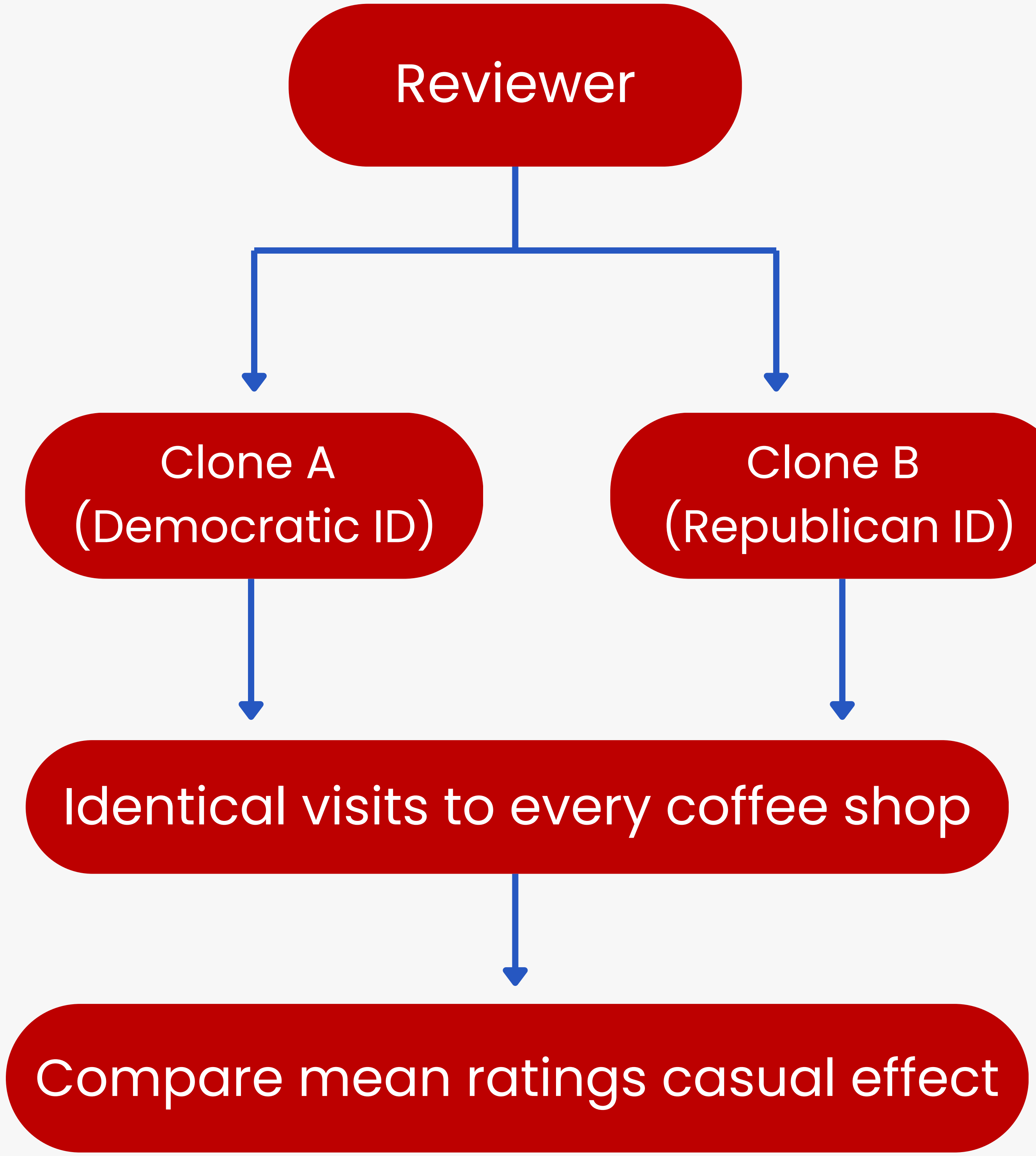
Regression Models
& Conclusion

Research Question

Do coffee shops located in Republican ("Red") cities receive higher ratings than those in Democratic ("Blue") cities?



Ideal Experiment



Ideal Design Logic

- Randomly assign political identity to reviewer clones
- Hold all non-political factors constant
- Same reviewer evaluates all coffee shops

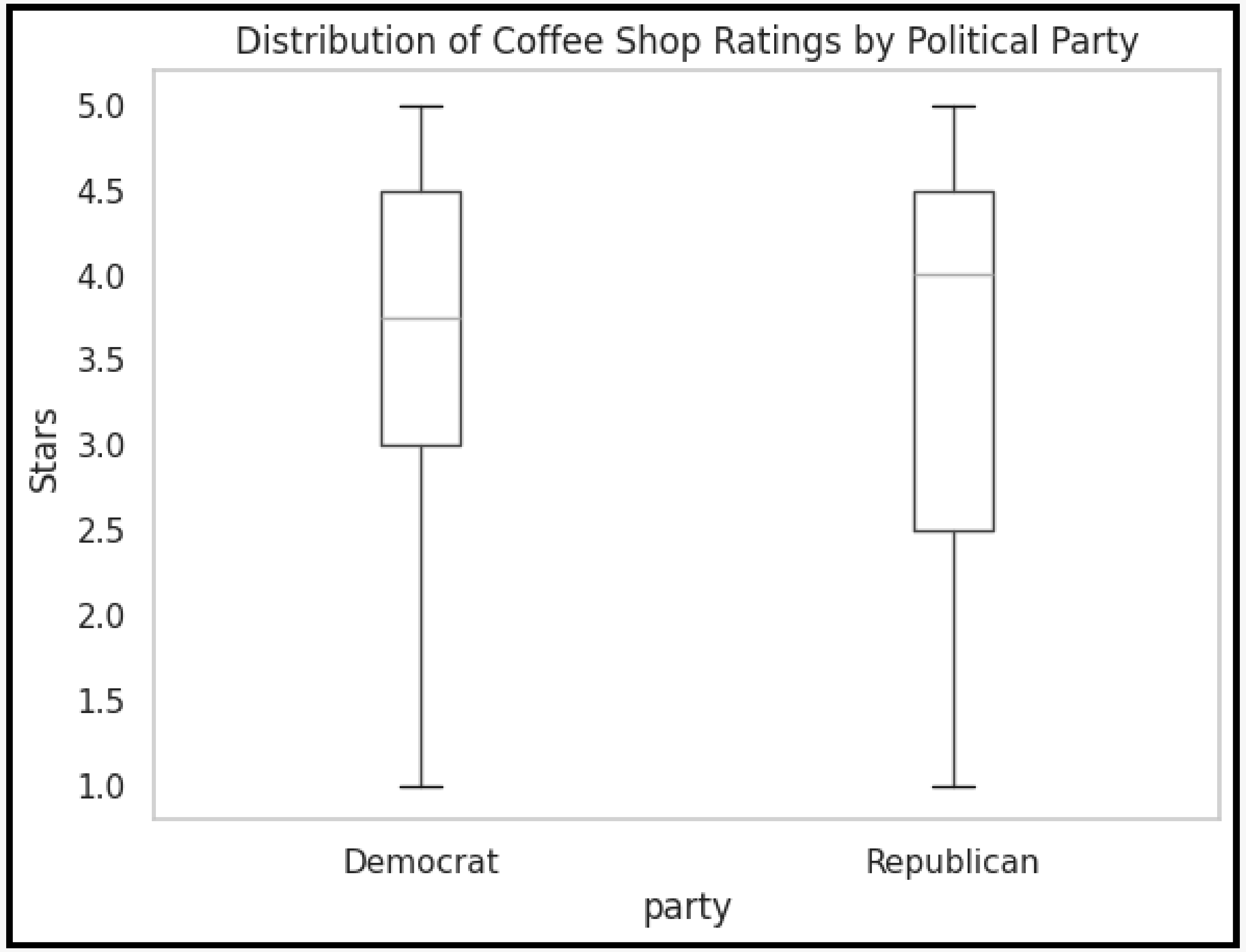
What the Ideal Experiment Shows

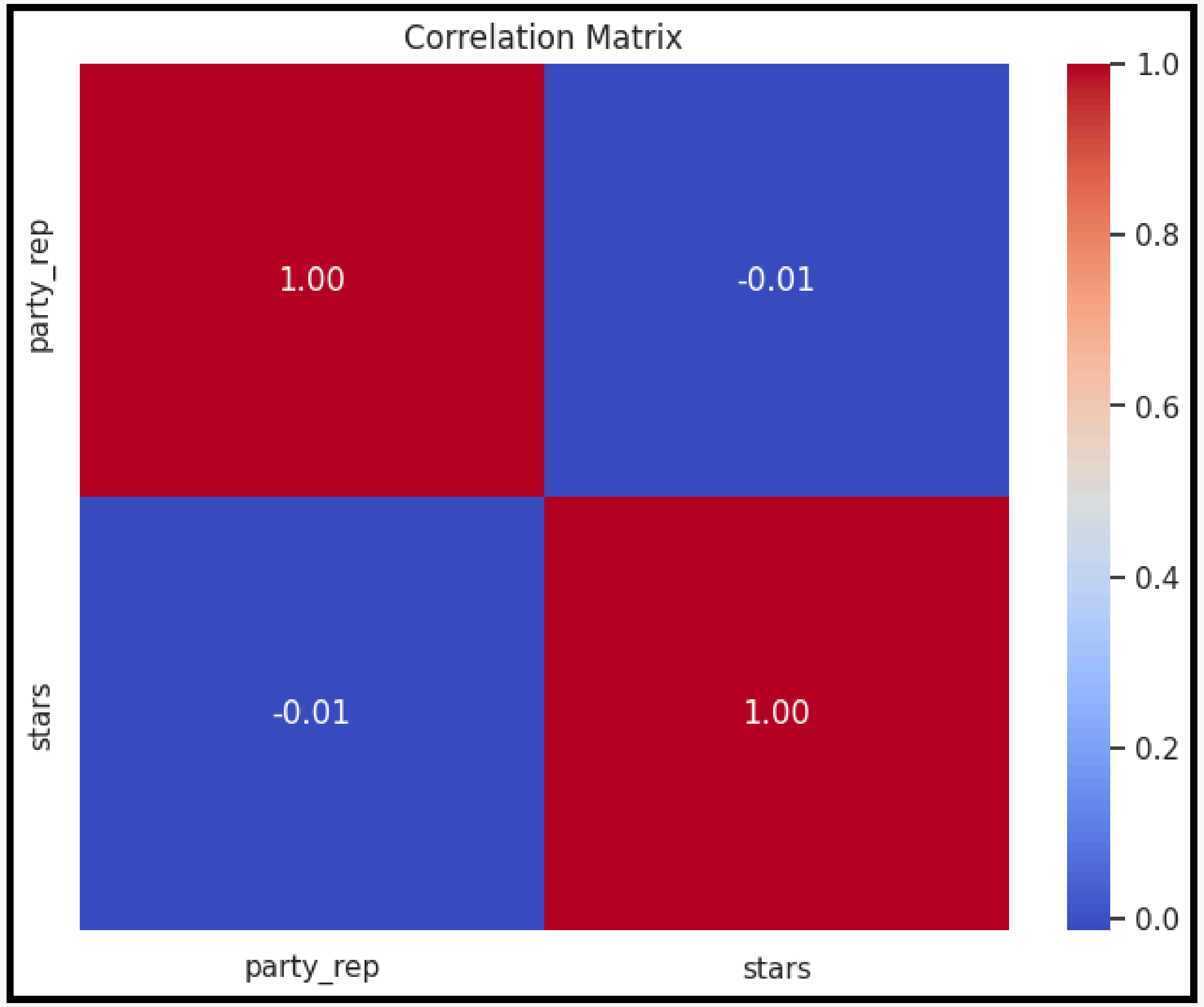
- Rating differences come from political identity only
- Causal effect = mean(Dem) - mean(Rep)
- No regression model needed in the ideal world

Why Reality Cannot Do This

- Cannot clone reviewers or standardize cities
- Cannot randomize real political environments
- Use controls + multivariable regression to approximate

Descriptive Statistics & Correlation Matrix





Mean rating for Democrat cities: 3.53
Mean rating for Republican cities: 3.5

Confounding Variable #1: Median Household Income

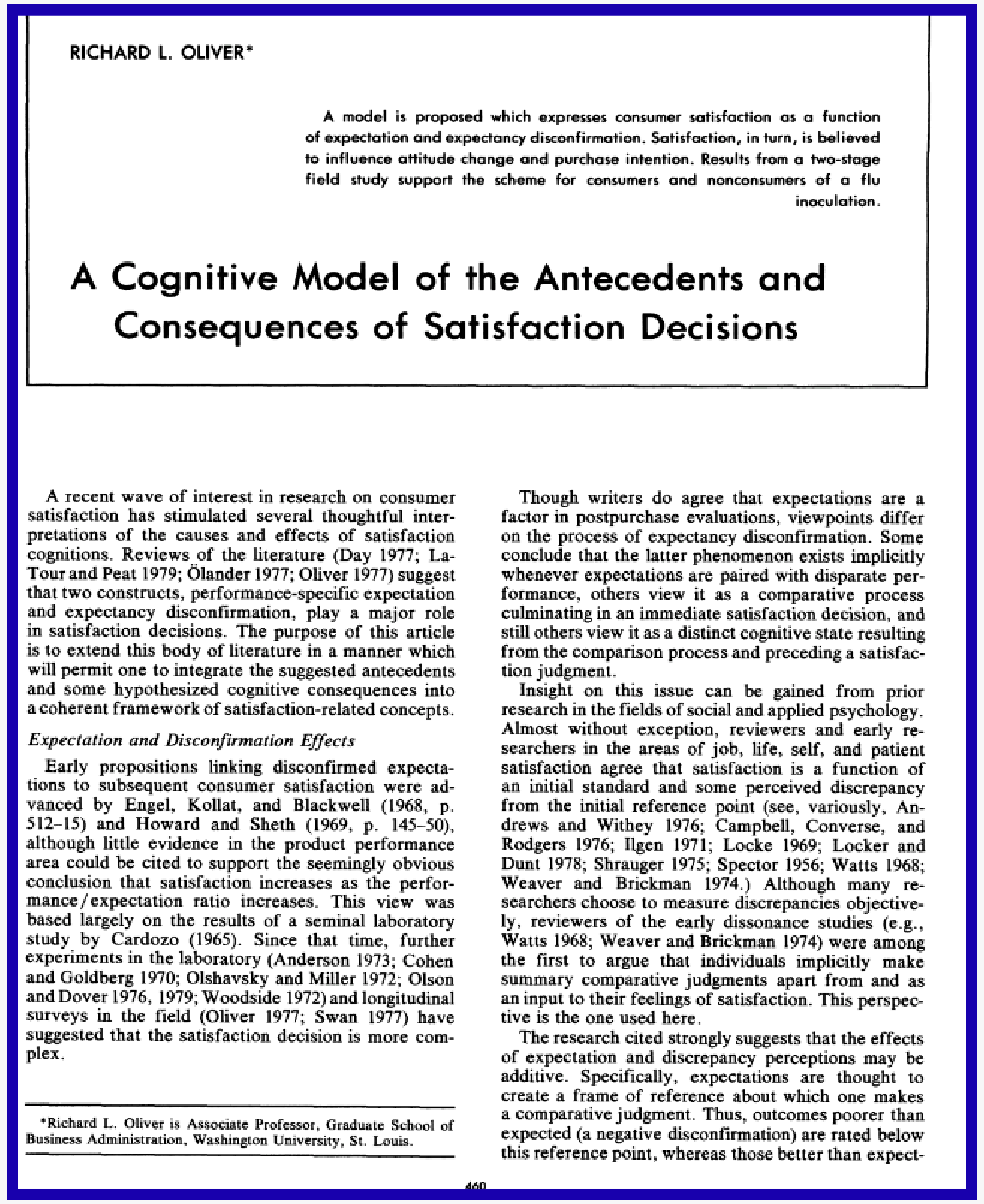
Theory: Expectation-Confirmation Theory (Oliver, 1980)

High SES (Socioeconomic Status)

↓

Higher Baseline Expectations

<u>Result:</u> Wealthier areas grade harsher for "average" service



RICHARD L. OLIVER*

A model is proposed which expresses consumer satisfaction as a function of expectation and expectancy disconfirmation. Satisfaction, in turn, is believed to influence attitude change and purchase intention. Results from a two-stage field study support the scheme for consumers and nonconsumers of a flu inoculation.

A Cognitive Model of the Antecedents and Consequences of Satisfaction Decisions

A recent wave of interest in research on consumer satisfaction has stimulated several thoughtful interpretations of the causes and effects of satisfaction cognitions. Reviews of the literature (Day 1977; LaTour and Peat 1979; Ölander 1977; Oliver 1977) suggest that two constructs, performance-specific expectation and expectancy disconfirmation, play a major role in satisfaction decisions. The purpose of this article is to extend this body of literature in a manner which will permit one to integrate the suggested antecedents and some hypothesized cognitive consequences into a coherent framework of satisfaction-related concepts.

Expectation and Disconfirmation Effects

Early propositions linking disconfirmed expectations to subsequent consumer satisfaction were advanced by Engel, Kollat, and Blackwell (1968, p. 512–15) and Howard and Sheth (1969, p. 145–50), although little evidence in the product performance area could be cited to support the seemingly obvious conclusion that satisfaction increases as the performance/expectation ratio increases. This view was based largely on the results of a seminal laboratory study by Cardozo (1965). Since that time, further experiments in the laboratory (Anderson 1973; Cohen and Goldberg 1970; Olshavsky and Miller 1972; Olson and Dover 1976, 1979; Woodside 1972) and longitudinal surveys in the field (Oliver 1977; Swan 1977) have suggested that the satisfaction decision is more complex.

Though writers do agree that expectations are a factor in postpurchase evaluations, viewpoints differ on the process of expectancy disconfirmation. Some conclude that the latter phenomenon exists implicitly whenever expectations are paired with disparate performance, others view it as a comparative process culminating in an immediate satisfaction decision, and still others view it as a distinct cognitive state resulting from the comparison process and preceding a satisfaction judgment.

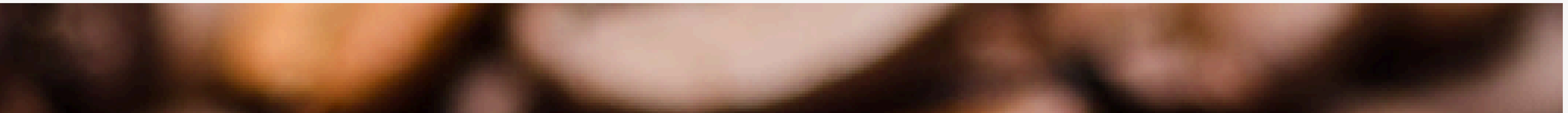
Insight on this issue can be gained from prior research in the fields of social and applied psychology. Almost without exception, reviewers and early researchers in the areas of job, life, self, and patient satisfaction agree that satisfaction is a function of an initial standard and some perceived discrepancy from the initial reference point (see, variously, Andrews and Withey 1976; Campbell, Converse, and Rodgers 1976; Ilgen 1971; Locke 1969; Locker and Dunt 1978; Shrauger 1975; Spector 1956; Watts 1968; Weaver and Brickman 1974.) Although many researchers choose to measure discrepancies objectively, reviewers of the early dissonance studies (e.g., Watts 1968; Weaver and Brickman 1974) were among the first to argue that individuals implicitly make summary comparative judgments apart from and as an input to their feelings of satisfaction. This perspective is the one used here.

The research cited strongly suggests that the effects of expectation and discrepancy perceptions may be additive. Specifically, expectations are thought to create a frame of reference about which one makes a comparative judgment. Thus, outcomes poorer than expected (a negative disconfirmation) are rated below this reference point, whereas those better than expect-

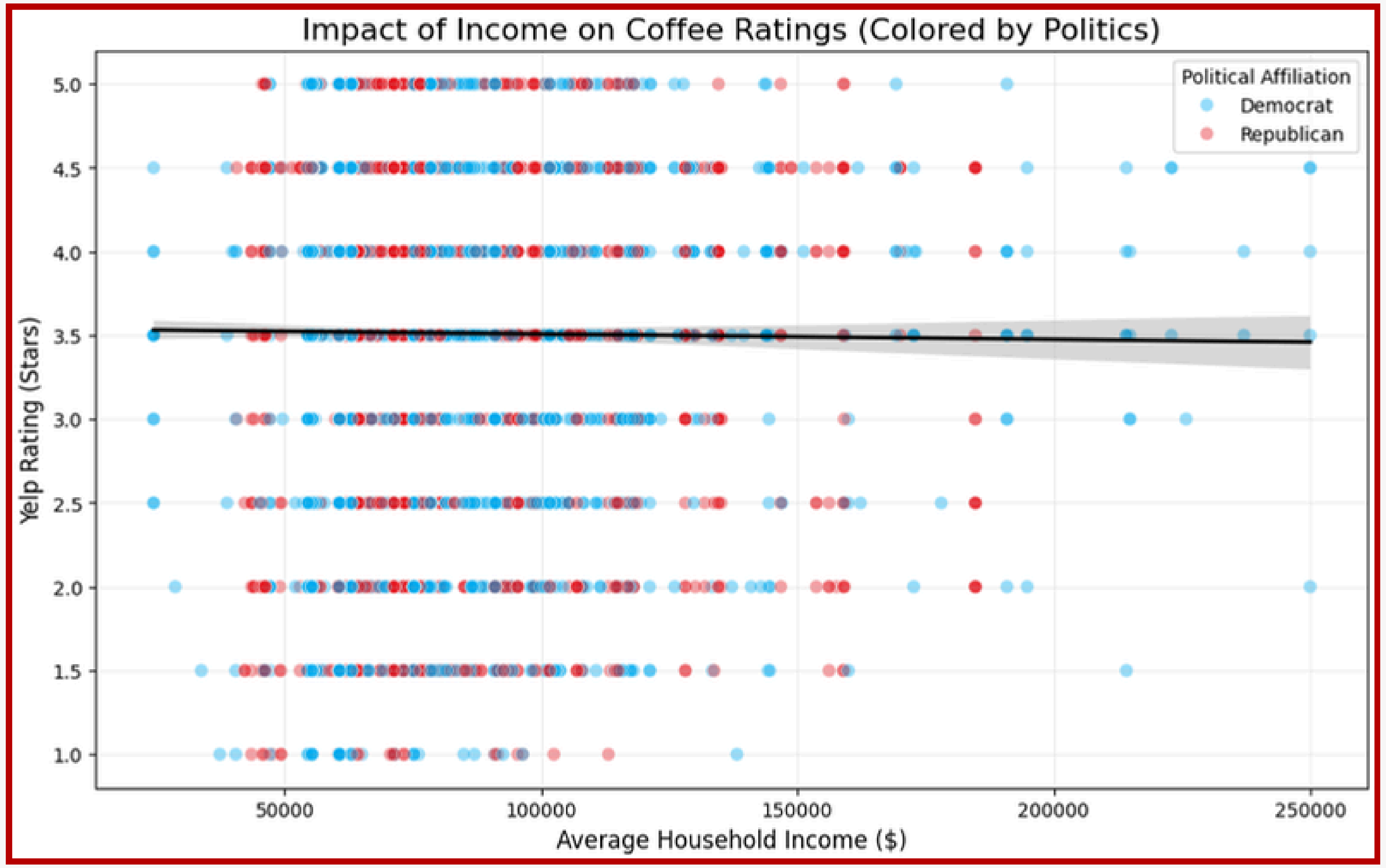
*Richard L. Oliver is Associate Professor, Graduate School of Business Administration, Washington University, St. Louis.

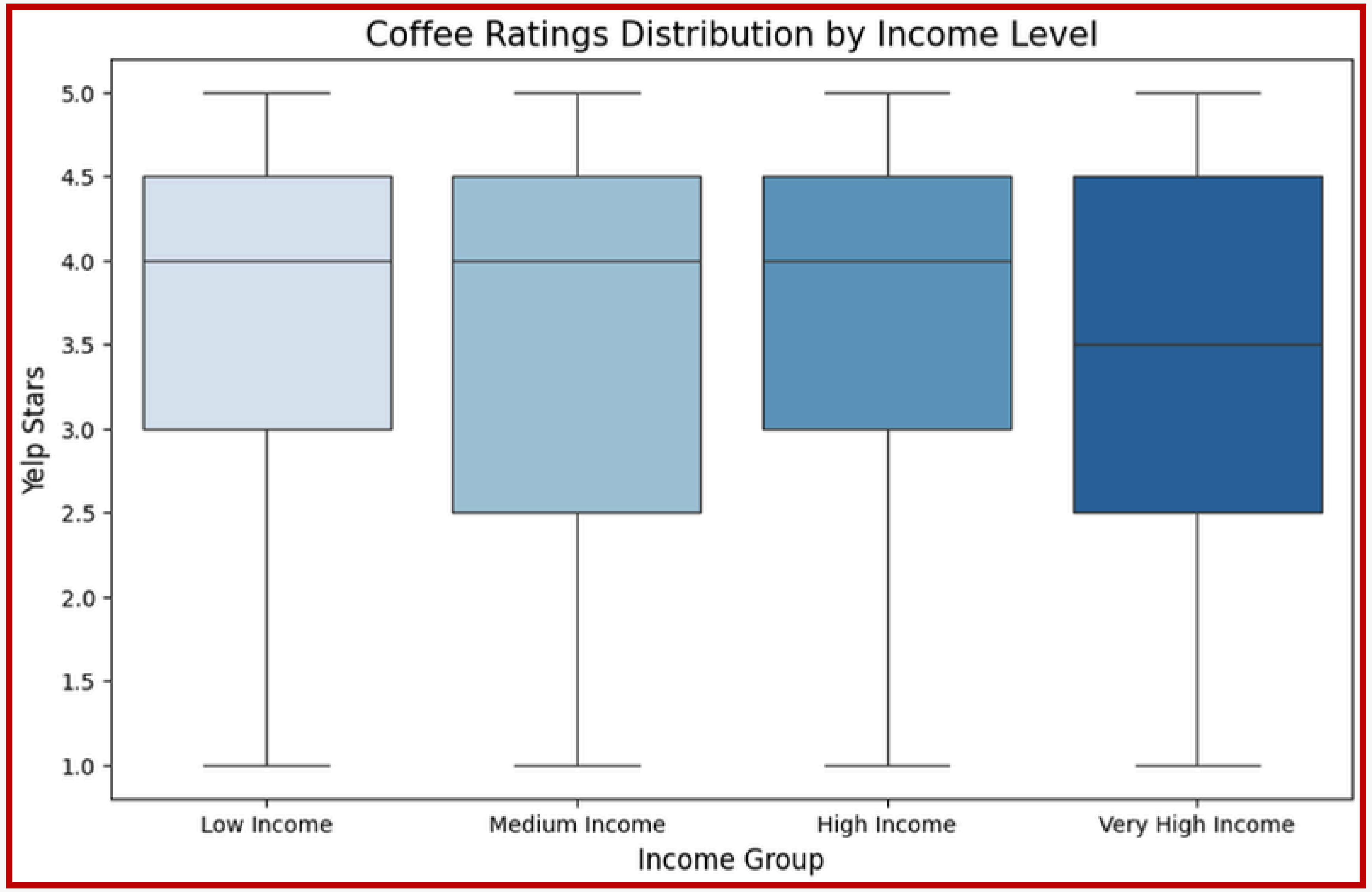
460

Source: Oliver, R. L. (1980). A Cognitive Model of the Antecedents and Consequences of Satisfaction Decisions. Journal of Marketing Research.



Median Income: Our Findings





Supports Theory: higher socioeconomic status drives higher standards and stricter grading

Confounding Variable #2: Education Level

No research on how general education level affects coffee shop rating, but the more a coffee shop engages and educates its customers, the higher the customer satisfaction, especially for specialty coffee.

For **Specialty** Coffee:

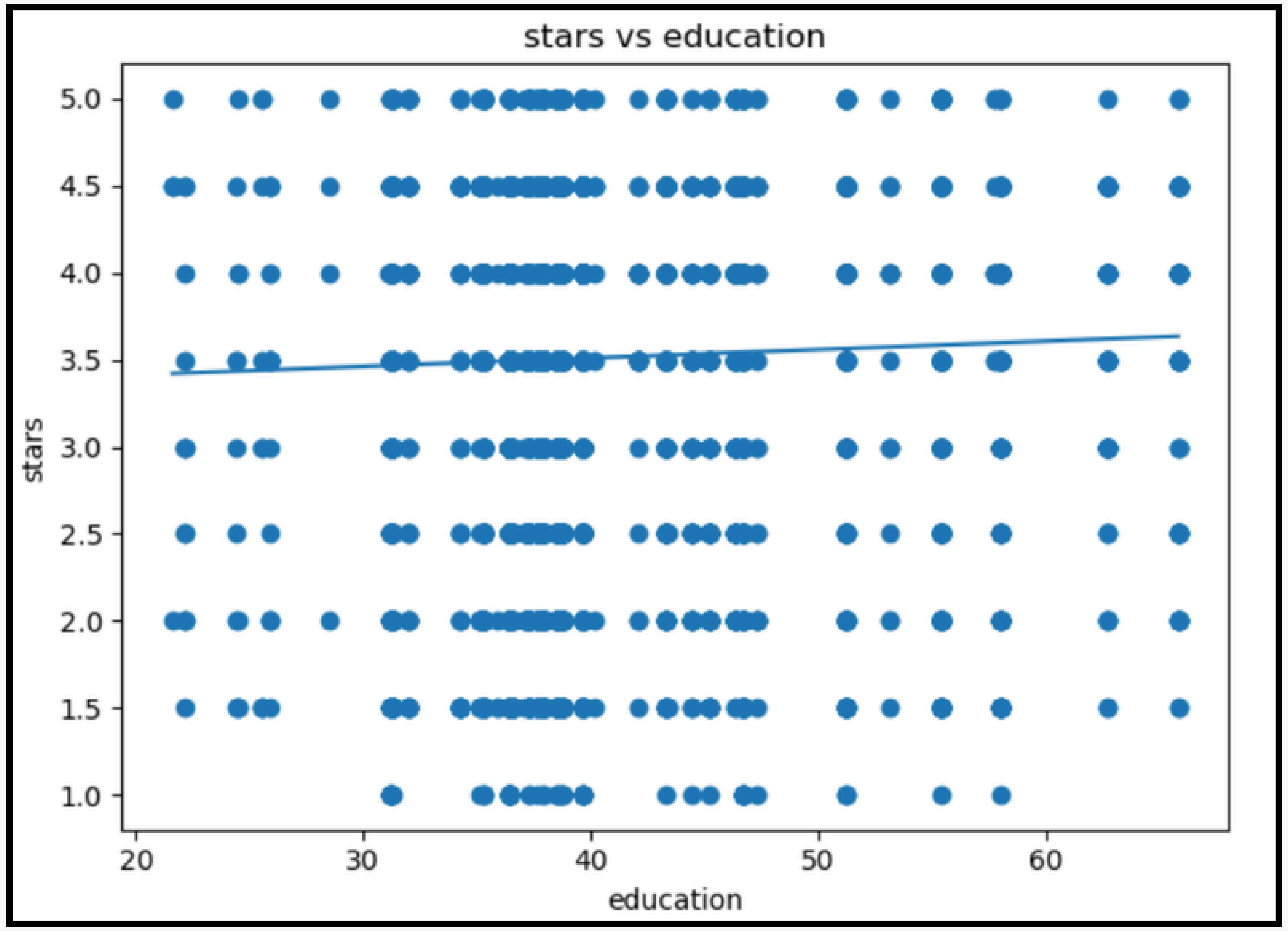
Knowledge of coffee

Satisfaction & engagement

**This may not be the case for regular coffee.*

Sources: Meeprom & Kokkhangplu (2025); Martinez & Ornelas (2021)

Education Level: Our Findings



No correlation between education and coffee shop rating

Confounding Variable #3: Median Rent

WHY RENT MIGHT MATTER

Design & Layout
Servicescape factors affect satisfaction

Cleanliness
Better environments → better customer experience

Coziness/Atmosphere
**Shen & Back (2016)*

Higher Rent

↓

Better Physical Enviornment

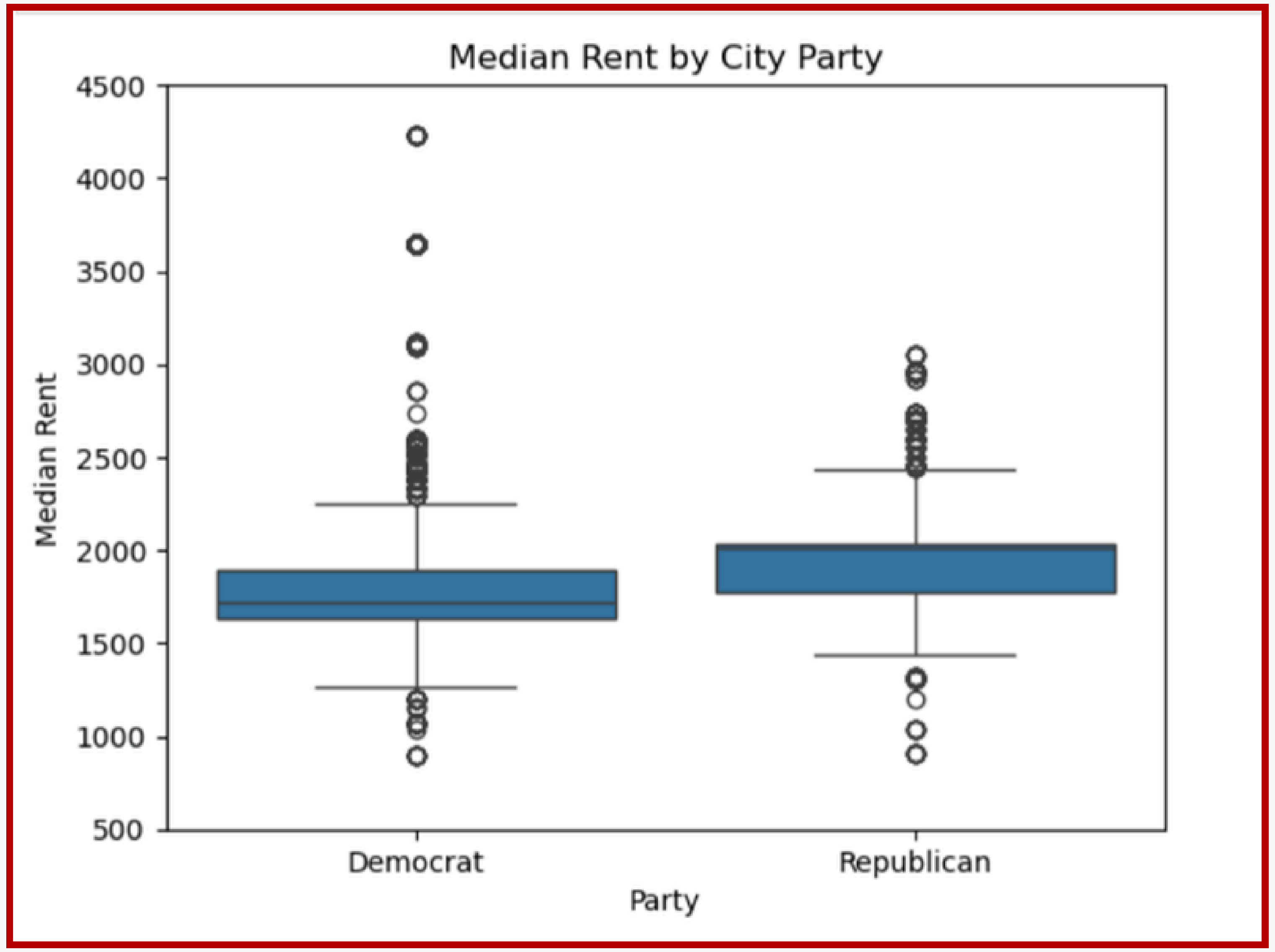
↓

Higher Customer Satisfaction

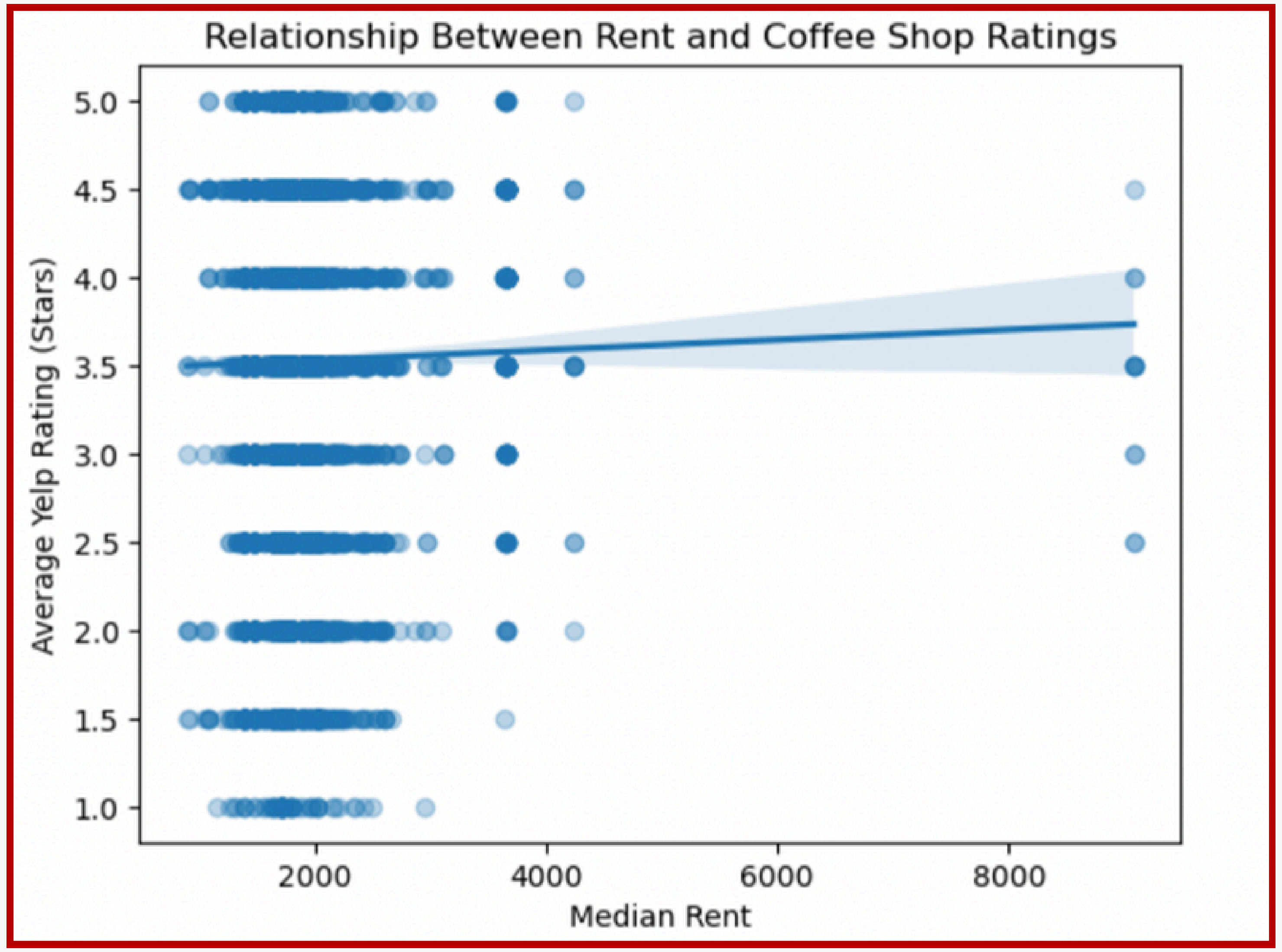
WHAT THIS MEANS FOR OUR STUDY

Shop in high rent cities may be able to invest in their physical space, which could contribute to slightly higher ratings. This creates a potential confounding effect when comparing Blue vs. Red cities.

Median Rent: Our Findings

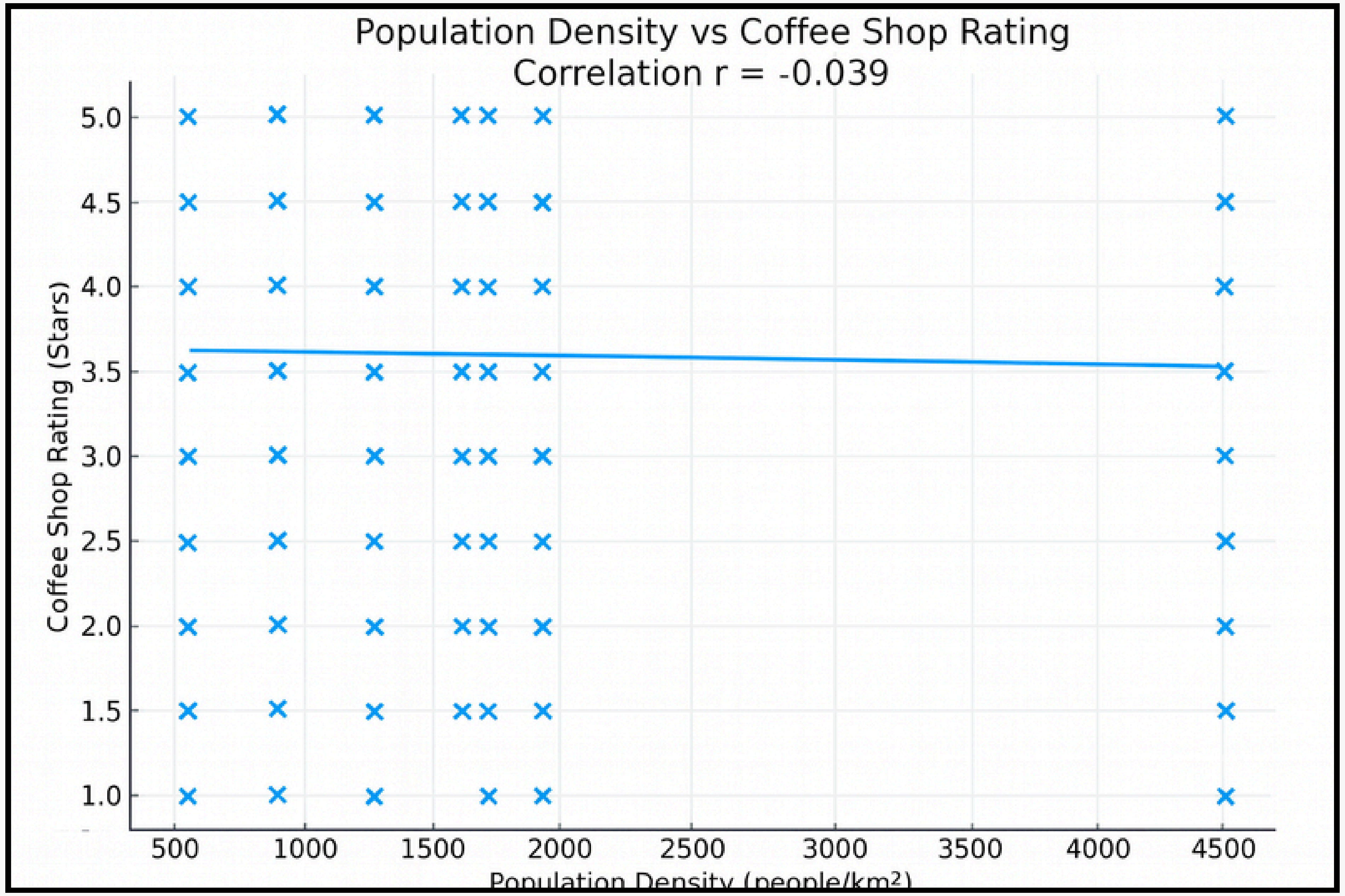


Democratic and Republican cities have similar median rent levels, but Democratic cities show more high-rent outliers. This suggests that rent differs more by city type than by political leaning.



The relationship between rent and coffee-shop ratings is nearly flat, indicating only a very weak positive association. Higher rent cities do not appear to systematically receive higher Yelp ratings.

Population Density



Population density shows almost no relationship with coffee shop ratings (r = −0.039).

Regression Result

Summary of Fit

RSquare	0.007992
RSquare Adj	0.006235
Root Mean Square Error	0.971345
Mean of Response	3.594026
Observations (or Sum Wgts)	2829

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	5	21.4589	4.29177	4.5487
Error	2823	2663.5302	0.94351	**Prob > F**
C. Total	2828	2684.9890		0.0004*

Lack Of Fit

Source	DF	Sum of Squares	Mean Square	F Ratio
Lack Of Fit	2	17.4083	8.70416	9.2794
Pure Error	2821	2646.1219	0.93801	**Prob > F**
Total Error	2823	2663.5302		<.0001*
				Max RSq
				0.0145

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|
Intercept	3.064691	0.248081	12.35	<.0001*
party[Democrat]	0.1331348	0.046002	2.89	0.0038*
education	0.0031466	0.005774	0.54	0.5858
salary	-1.145e-5	4.056e-6	-2.82	0.0048*
median_rent_2025_10_31	0.0006769	0.000242	2.80	0.0052*
population_density	-0.000052	0.000019	-2.75	0.0060*

Democratic counties are rated about 0.13 stars higher than those in Republican counties.

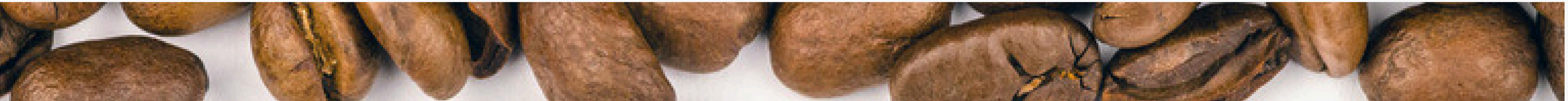
Higher salary and population density are linked to slightly lower ratings, and higher median rent is linked to slightly higher ratings.

P value for F test is 0.0004, but RSquare is 0.008.

One Observation

Predicted stars	Lower 95% Indiv stars	Upper 95% Indiv stars
3.543192284	1.6375111158	5.4488734522

For St Honore Pastries in Philadelphia (Democratic county with 36.4% college graduates, median rent $1,715, income $60,698, and population density 4,517), the model predicts an average rating of 3.54 stars, with a 95% prediction interval from 1.64 to 5.45 stars.



Thank You So Much!

Any Questions?

Presented by **Team B8**